VISUAL GROUNDING × IDENTITY-CONSISTENT REASONING

RoboPIN

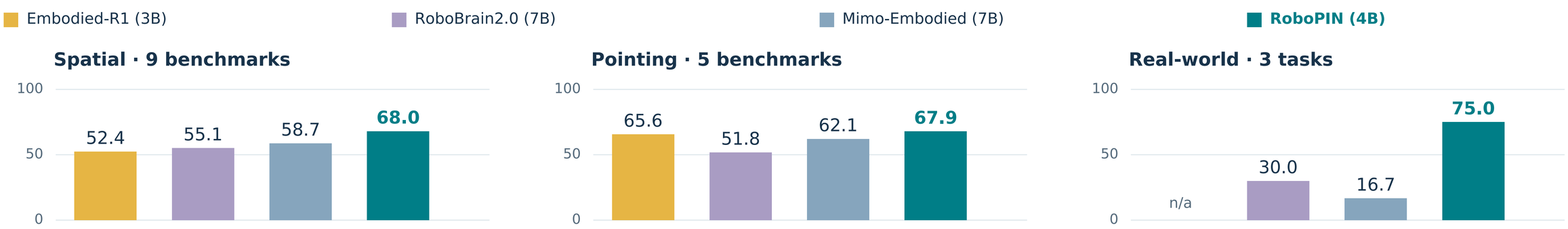
Grounded Embodied Reasoning via Pinned Chain-of-Thought

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Tianjin University | ACM Multimedia 2026 · Rio de Janeiro, Brazil

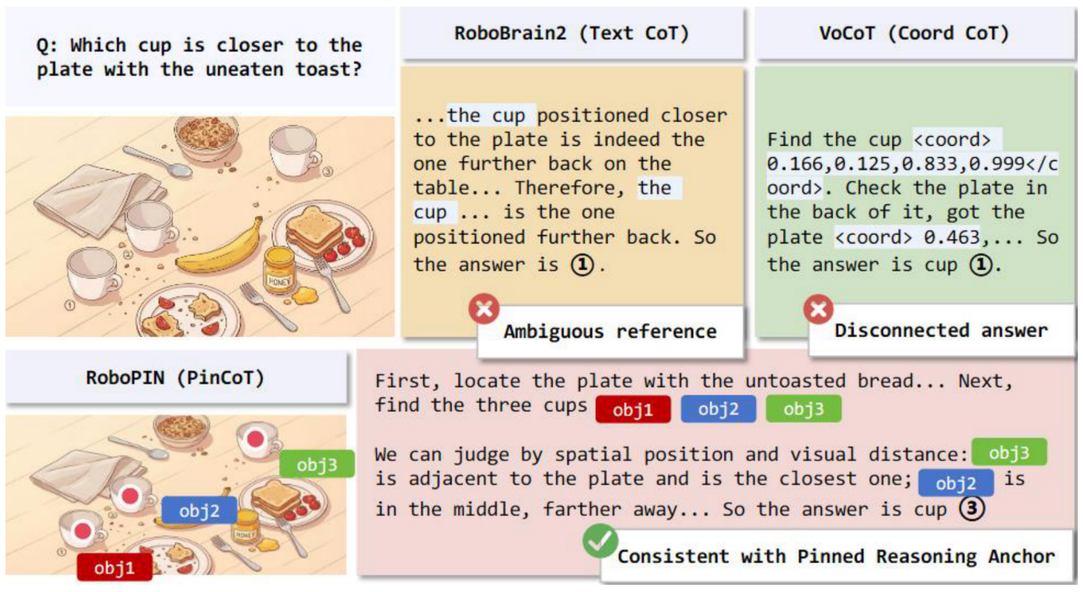
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Overall performance across task families



Scores (%), Tables 1–3 / Figure 1. Spatial: 13-column mean; pointing: 5-benchmark mean; robot: success over 60 trials.

01 One entity. One persistent identity.



Ground reasoning in persistent visual anchors.

Text CoT

Implicit entity references can drift.

Coordinate CoT

Spatial coordinates alone do not preserve identity.

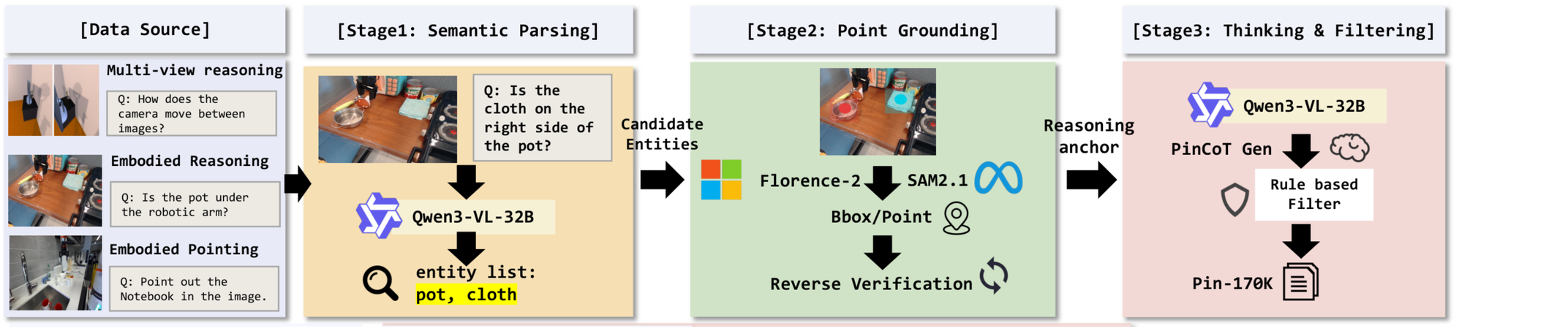
PinCoT / ours

Bind every step to an explicit, trackable entity.

Name + ID + view + point

Original comparison: paper Figure 1 (left).

02 From data to RoboPIN



Three-stage post-training

SFT — Embodied knowledge → **CoT-SFT** — Structured reasoning → **RFT / GRPO** — Process alignment

03 Results & process analysis

RoboPIN 4B vs. Mimo-Embodied 7B

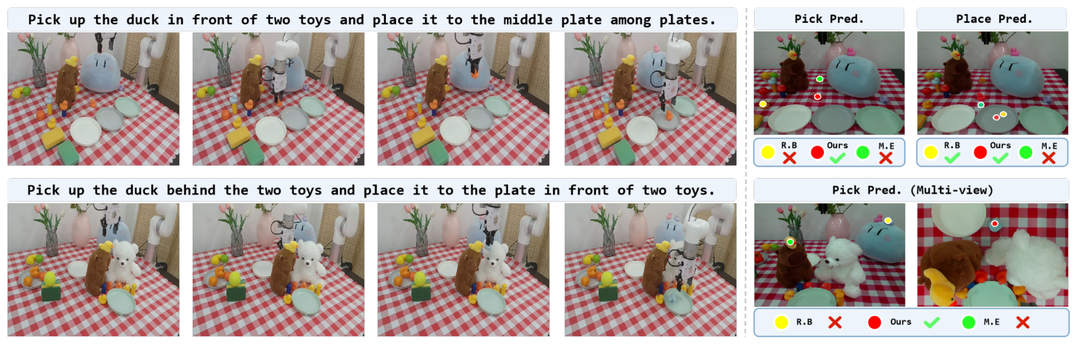
Benchmark	7B	4B / ours	Δ pp
EmbSpatial	76.2	**83.1**	**+6.9**
RoboSpatial	61.8	**66.6**	**+4.8**
CrossPoint	36.8	**72.2**	**+35.4**
BLINK · multi-view	26.3	**72.9**	**+46.6**
RoboRefit	82.3	**89.0**	**+6.7**
RoboAfford	69.8	**72.8**	**+3.0**

Selected results, Tables 1–2; Δ in percentage points.

Cross-step ID coverage: 18.8% → 99.9%
Without RFT → RoboPIN · EmbSpatial, Table 7

04 Zero-shot robot transfer

xArm · static + wrist cameras · Figure 3 / Table 3



Successful executions / 20 per task

20 / 20	14 / 20	11 / 20
Spatial	Multi-view	Long-horizon

Ground every step. Preserve every identity.

PinCoT makes visual evidence trackable across reasoning steps and viewpoints, supporting reliable embodied decisions.

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